SURFACE PROXY · SPATIAL ANALYSIS · OPEN WORKFLOW

Surface titanium and lunar crustal magnetism

Global-map analysis over the declared joint-valid footprint (LROC ±70° and Imbrian-age restriction). **It does not directly test** the dated temporal mechanism of Nichols et al. (2026).

lemoon01110 · github.com/lemoon01110/Lunar-Titanium-Magnetism

EDUCATIONAL & AI-ASSISTED

This presentation and the accompanying repository are a personal project created for learning and hands-on experience.

Artificial Intelligence (Large Language Models) was used extensively throughout its development to help write the code, understand the geophysical concepts, and rigorously catch statistical and logical errors.

THE PUZZLE

A magnetized crust with no magnet behind it

- **No global field today**

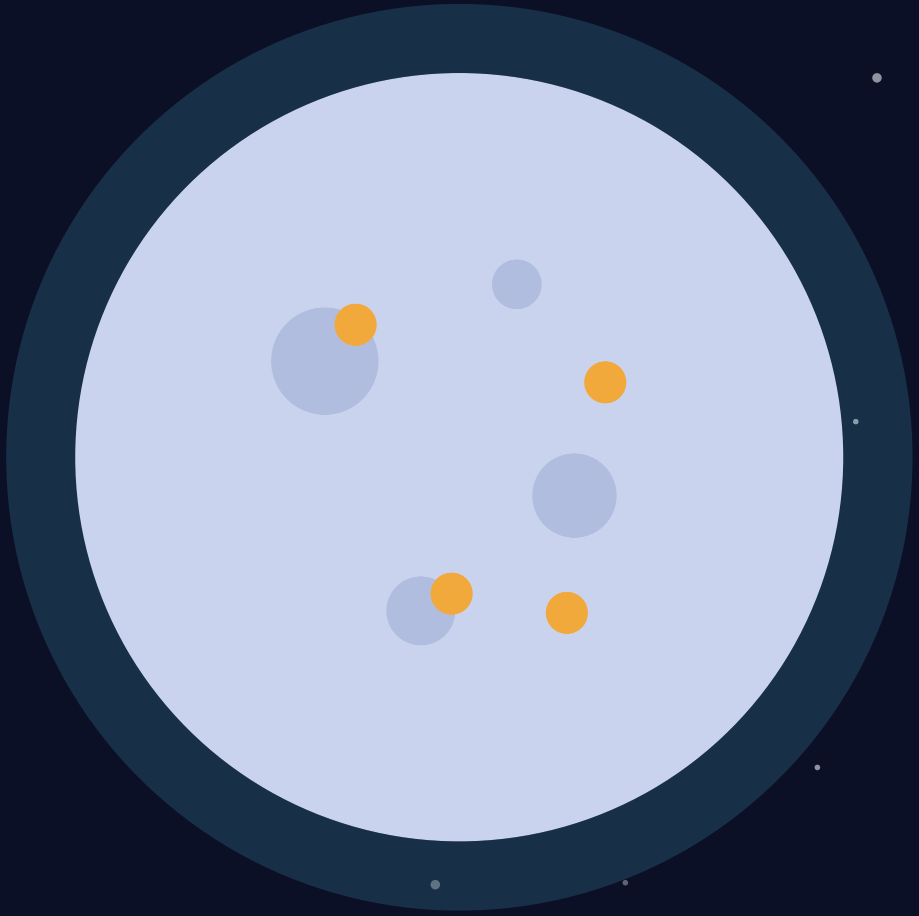
  Unlike Earth, the Moon has no active core dynamo — so it should be magnetically dead.

- **Yet the crust is magnetized**

  Orbiting spacecraft measure strong, patchy magnetic anomalies locked into ancient rock.

- **Timing is the missing variable**

  Dated samples constrain ancient field evolution. Present-day surface maps test spatial co-location, not the temporal dynamo mechanism.



THE SCOPE

One temporal mechanism motivates one spatial proxy test

Motivation · dated samples

Nichols, Wade & Stephenson (2026), Nature Geoscience

Dated palaeointensity and Ti-bearing samples motivate a time-dependent core–mantle boundary melting and dynamo hypothesis.

Temporal mechanism

Benchmark · impact antipodes

Hood & Artemieva (2008), Icarus

Fold-matched H2 PR-AUC .110579 vs null mean .031970, SD .026728 and 95th .106832 (p=.039604).

100 accepted shifts · 5/5 folds evaluable

THE ANSWERABLE QUESTION

Does surface TiO_2 add spatial information for crustal magnetism after mapped geologic controls?

1

Repository plan

Criteria were committed in repository history, but there is no independent external timestamp.

2

Power-gated

Positive and negative outcomes are explicit; negative claims require adequate detection power.

3

Scope-limited

The result concerns spatial co-location—not Nichols et al.'s temporal dynamo mechanism.

THE EVIDENCE

Five global datasets, plus a mapped mare-proxy mask

- **Magnetic field (what we predict)**
  Kaguya + Lunar Prospector surface vector map
- **Surface titanium — TiO_2 (proxy)**
  LROC WAC UV/Vis abundance map; mare-derived calibration
- **Bouguer gravity**
  GRAIL GRGM1200A — subsurface density structure
- **Crustal thickness**
  GRAIL crustal-thickness archive
- **Geology + mare proxy**
  USGS units define age and mapped mare-proxy scope

~2 GB
of institutional data

23
pinned data products

SHA-256
every byte verified

1° grid
≈ 30 km at the equator

THE METHOD

Four stages separate description from inference

1

Align

Align five maps and the derived mapped mare-proxy mask on one equal-area lunar grid.

2

Describe

Fit continuous amplitude with controls, then add Ti features: raw + 25/50/100 km.

3

Restrict

Repeat controls vs. controls+raw-TiO_2 only inside the mapped USGS mare-proxy mask.

4

Calibrate

Match each statistic to its own spatial null, then inject H2 signals to measure power.

Why blocking still falls short: 30° (~910 km) and 60° (~1819 km) holdouts remain below the ~3752 km dependence range. Both nulls used the first 100 eligible 5/5-fold unique nonidentity shifts from 125 candidates; 12 four-fold candidates were rejected. Accepted prevalence: .007686–.022767.

AUDITABILITY

Reproducible evidence still has design limits

Repository self-attestation

The plan predates the run in git history, but lacks an independent external timestamp.

Logical-fallacy audit

Inference traps are documented and corrected in the open—including the matched H2 null.

Fail-closed provenance

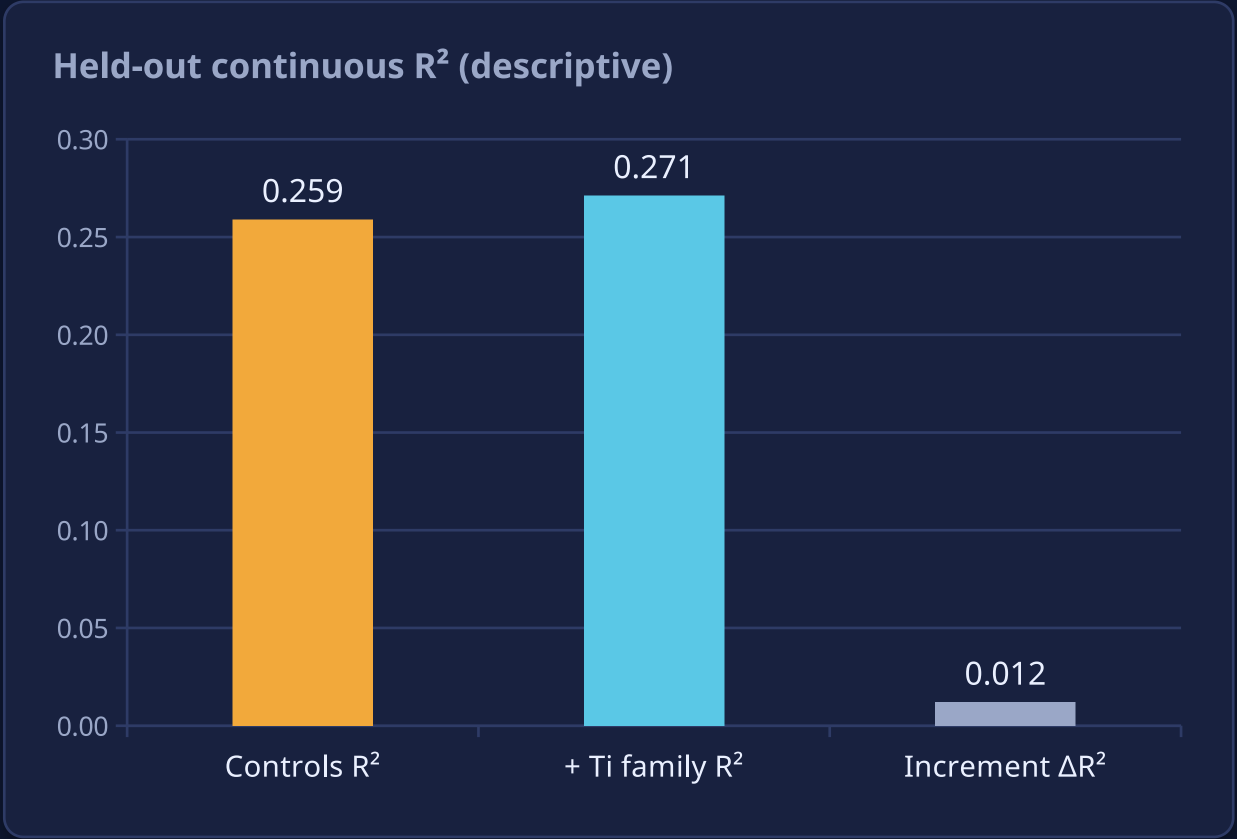
Hashes, real/synthetic guards and mare-mask alignment are enforced in code.

Automated tests

Guards cover provenance, terrain validity, power gating and report claims.

THE RESULT

The full Ti family adds only a small descriptive increment



ΔR^2 +.0121

controls .2590 → .2712 with full Ti family; descriptive only

ΔPR-AUC −.002992

remove raw/buffered Ti + Ti×gravity interactions; p=.78125

n_eff ≈ 1

30° 910 km holdout < 3752 km dependence range

STATUS: INCONCLUSIVE_LOW_POWER

This spatial proxy result neither confirms nor refutes Nichols et al.'s temporal mechanism.

TERRAIN VALIDITY

Mapped mare-proxy TiO_2 is suggestive but unstable

Mapped mare-proxy binary

ΔPR-AUC +.067614; +TiO_2 .125209 versus controls .057595 (p=.21875).

n=6,232 · 58 positives

Mapped mare-proxy continuous

Controls R^2 .425691 → .438387; raw-Ti increment +.012696±.012575. Common-fold gain vs full raw-Ti: +.001445.

proxy range 3289 km · n_eff=1

Interpretation: Both proxy checks remain descriptive: the mapped proxy footprint still supplies only one dependence-scale unit, not **an independent global confirmation.**

Three diagnostics set the inference ceiling

Dependence exceeds blocks

n_eff≈ 1. Even 60° blocks (~1819 km) remain below the ~3752 km dependence range.

Matched H2 benchmark recovered

Full PR-AUC .088972 vs fold-matched null mean .112697, SD .056903 and 95th .225555 (p=.564356); H2 match p=.039604.

Power floor is not adequate power

H2 recovery at β=0/.5/1/1.5/2/3–4: 3/30, 18/30, 27/30, 28/30, 29/30, 30/30. Point MDE β=1; Wilson-lower MDE β=2.

“

INCONCLUSIVE
LOW POWER

β=1 implies risk +.0557 and OR≈607; no justified target effect.

The next test must be time-resolved.

- Treat this as a surface-proxy spatial study—not a temporal dynamo test.
- Use dated samples or time-resolved proxies to test the Nichols mechanism directly.
- Expand calibrated TiO_2 coverage and use dependence-scale blocks before any negative claim.

Thank you · github.com/lemoon01110/Lunar-Titanium-Magnetism